Machine Learning for Finance

Unsupervised Learning - Clustering

Prof. Mike Aguilar
Economics Department and Fuqua School of Business, Duke University

https:/www.linkedin.com/in/mike-aguilar-econ

Outline

1. Clustering
   - K-Means
   - Gaussian Mixture Modeling
   - Hierarchical
   - Minimum Spanning Tree

Motivation

- These notes accompany the Duke Economics course *Machine Learning for Finance*.
- The course focuses on how machine learning methods can be used to design, evaluate, and implement systematic investment strategies.
- Our goal is to introduce a range of ML techniques with enough mathematical structure to build intuition, while avoiding unnecessary formalism.
- Throughout the notes we use a simple return–momentum signal as a running example. Although momentum is not always the ideal illustration for every technique, using a single, familiar signal provides a consistent thread. This is especially true when we transition to real data and empirical implementation.
- The methods we discuss are broadly applicable across many signals, asset classes, and investment settings.

Description of ML Methods

We'll use a toy momentum example as motivation

Unsupervised Learning

- **Clustering**
  Group similar assets by momentum characteristics
- **Dimension Reduction**
  Condense multiple momentum measures into fewer factors
- **Generative Modeling**
  Simulate returns to derive momentum structure

Supervised Learning

- **Classification**
  Predict positive/negative returns using momentum
- **Regression**
  Predict cross-sectional returns using momentum inputs

Time Series Modeling

- **Unsupervised**
  Group assets by temporal behavior, then measure momentum
- **Supervised**
  Forecast returns based on persistence

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Clustering Momentum

Q: How do we form a momentum signal using any clustering method?

1. Convert prices to returns, then compute momentum
2. Group assets by momentum using a clustering method (e.g., K-Means, GMM, hierarchical). Each asset receives a cluster label based on the method's assignment rule.
3. Form a long/short signal by identifying:
   - the cluster with the highest average momentum (the "winner" group), and
   - the cluster with the lowest average momentum (the "loser" group),

   and going long assets in the winner cluster and short assets in the loser cluster.

Types of Clustering

- Hard: Assigns each observation to a single cluster
- Soft: Assigns each observation to potentially more than one cluster with a degree of "belonging"

Clustering Methods

1. Centroid-based — Clusters by grouping observations around representative feature coordinates (e.g., K-Means, K-Medians).
2. Model-based — Clusters by assuming data arise from overlapping probability distributions and assigning observations by likelihood (e.g., Gaussian Mixtures).
3. Hierarchical — Clusters by building a tree of nested groups formed through successive merges or splits.
4. Graph-based — Clusters by connecting observations through a minimum spanning tree and cutting long edges to reveal groups.
5. Density-based — Clusters by identifying dense regions of observations separated by sparse areas (e.g., DBSCAN).

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K-Means

The most common Centroid Based, Hard Clustering method

1. Choose K as the number of clusters
2. Assign each observation x_i to one of the K clusters
3. Compute the mean (aka centroid) of each cluster $\bar{x}^k$
4. For each observation
   - Update cluster assignments by minimizing the distance between each observation and its cluster centroid (i.e. it's cluster "belonging")
   - Update cluster means by recomputing $\bar{x}^k$ for each cluster
   - Repeat until stopping criterion is met

Output

- Cluster means $\bar{x}^k$
- Cluster assignments x_i^k for $k = 1, \cdots, K$; $i = 1, \cdots, N$
- Clustering Error (Within Cluster Sum of Squares) $\frac{1}{N} \sum_{i=1}^{N} (x_i^k - \bar{x}^k)^2$

Considerations for Implementing K-Means

1. Accommodating multidimensional x
2. Choosing the number of clusters
3. Initializing cluster assignments
4. Pre-processing — helpful to standardize, or otherwise scale, the data before clustering, especially when using Euclidean distance
5. Tie Breaking
6. Empty Clusters
7. Stopping Criterion

Note: We won't dwell on 5,6,7 in the list above.

Multidimensional x

Suppose $\mathbf{x}_i^k$ is not a scalar; e.g. Clustering momentum AND volume across the SP500, in which case $dim(\mathbf{x}_i) = 2 \times 1$

- Cluster centroids are columnwise; i.e. (avg momentum, avg volume)
- Clustering Error expands to $\frac{1}{N} \sum_{d=1}^{2} \sum_{i=1}^{N} (x_{i,d}^k - \bar{x}_d^k)^2$

Ticker	Momentum	Errors	Squared Errors
A	1	-1.5	2.25
B	2	-0.5	0.25
C	3	0.5	0.25
D	4	1.5	2.25
Avg	2.5		1.25
Sum			5

Ticker	Volume	Errors	Squared Errors
A	200	-200	40000
B	300	-100	10000
C	700	300	90000
D	400	0	0
Avg	400		35000
Sum			140000

			Error		Squared Errors		
Ticker	Momentum	Volume	Momentum	Volume	Momentum	Volume	Squared Errors
A	1	200	-1.5	-200	2.25	40000	40002.25
B	2	300	-0.5	-100	0.25	10000	10000.25
C	3	700	0.5	300	0.25	90000	90000.25
D	4	400	1.5	0	2.25	0	2.25
Avg	2.5	400					35001.25

Sum of Avg	35001.25
(1/4)*Sum of Sums	35001.25

Note: Euclidean distance is not the only distance measure available for K-Means

Choosing the number of clusters

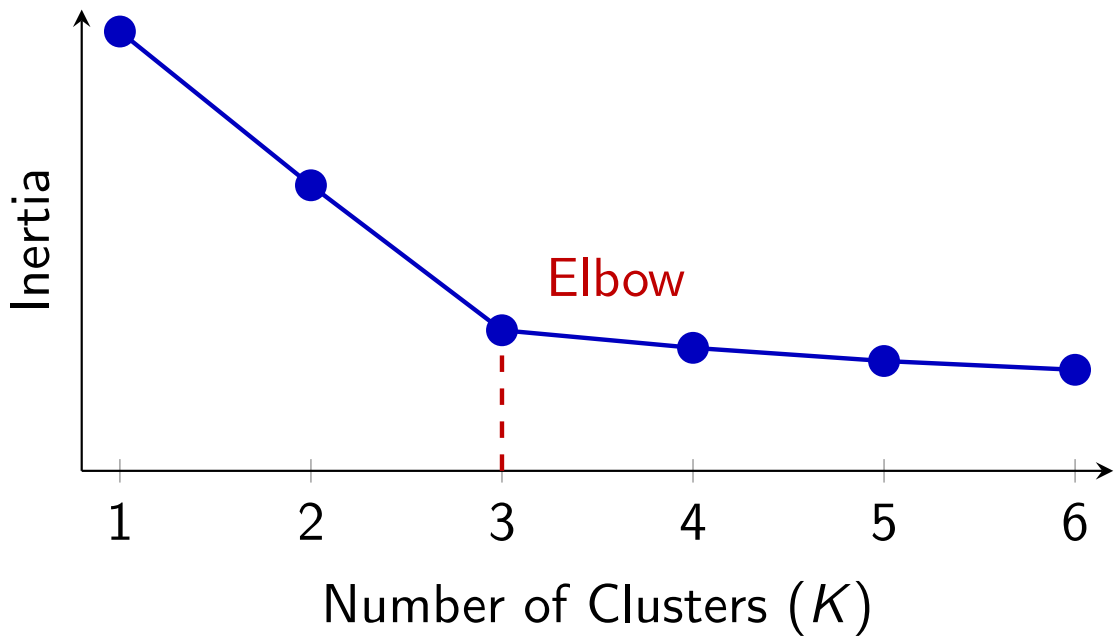
- If we set $K = 1$, then we are basing analysis on the single observation that has the smallest distance measure (e.g. the "average" observation)
- If we set $K = N$, then each observation is its own cluster

Typical Approaches

1. Choose K based on context; e.g. Long, Neutral, Short $K = 3$
2. If clustering is an aspect of our feature engineering to a supervised ML exercise, we can chose the K that optimizes the accompanying supervised validation error
3. If clustering is primarily part of EDA, then we can use "elbow method". Plot the within Cluster Error (aka Inertia) for various k and chose the K where the rate of decrease in the error slows.

Choosing Number of Clusters with Elbow Method

Q: How many clusters are suggested?



A: The elbow occurs at $K = 3$, where the curve changes from a steep decline to a relatively flat slope.

- Inertia measures within-cluster variation (within cluster sum of squares); lower values indicate tighter clusters.
- Adding clusters always reduces (or leaves unchanged) inertia, but improvements diminish beyond the elbow.
- The elbow is a subjective heuristic, not a formally optimal choice of K.

Initializing Cluster Assignments

A few of the many approaches available

- Randomly, assuming a uniform distribution, choose rows from the dataset as the initial centroids.
- (K-means ++) Ensure the initial centroids are well spread out
  1. Pick first centroid randomly
  2. Compute squared distances of each observation to that centroid
  3. Choose next centroid with probability proportional to the squared distance so that points farther away are more likely to be chosen
  4. Repeat until K centroids are chosen.

Judging Cluster Assignment with Silhouette Plots

Silhouette Score

$$s_i = \frac{b_i - a_i}{max\{a_i, b_i\}} \in \{-1, \cdots, +1\}$$

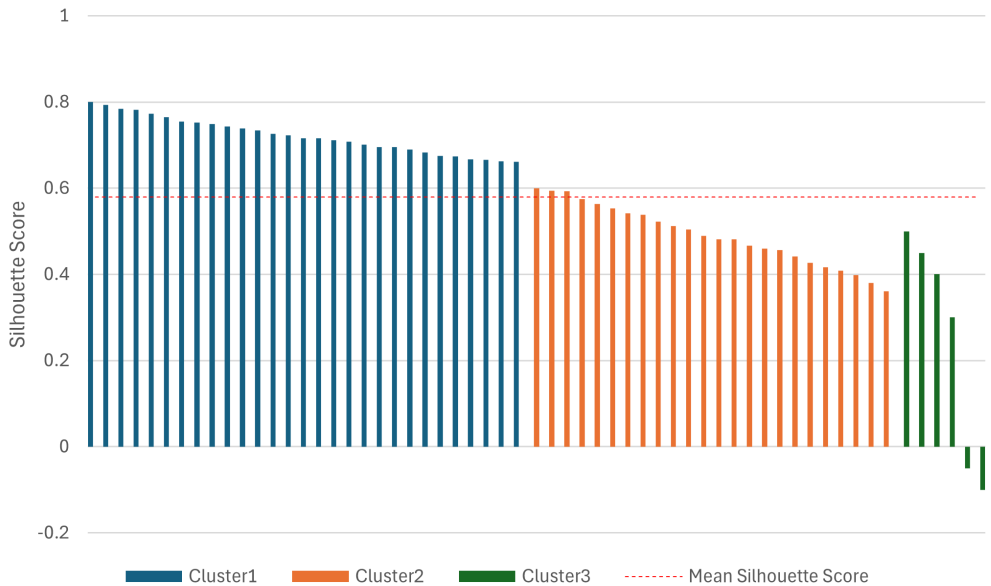
where

- a_i : cluster cohesion; average distance of observation i to all other points in its own cluster
- b_i : cluster separation; average distance between observation i and all observations in the nearest cluster (to which i is not assigned)

We compute the mean Silhouette score $\bar{s}$ and compare the individual scores to that mean. If the scores within a cluster are less than the $\bar{s}$, it implies that this cluster might be too dispersed (large a_i) and/or too close to other clusters (small b_i).

Good and Bad Clusters

Q: Which cluster(s) are good/bad? What does a negative score mean?



A:

- Cluster 3 seems "bad". Many of the observations assigned to this cluster have lower scores than average.
- The negative scores imply that these observations are actually closer to another cluster than to the cluster to which they are assigned.

K-Means Clustering Momentum

Q: How form a momentum signal via K-Means Clustering?

1. Choose the number of clusters K via Elbow, Silhouette, etc.
2. Group assets by momentum via K-Means. Each asset is assigned to the cluster whose centroid is closest to its momentum value.

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

$$p^{(+)} = \mathbf{1}\left[k_t^i = k_t^{(+)}\right], \qquad p^{(-)} = \mathbf{1}\left[k_t^i = k_t^{(-)}\right]$$

where

- $k_t^i = \arg\min_{k \in \{1,\ldots,K\}} \|X_{i,t} - \mu_{k,t}\|$ (label of nearest cluster, not the distance)
- $\mu_{k,t} = \frac{1}{N_{k,t}} \sum_{i:k_t^i = k} X_{i,t}$ (centroid of cluster k at time t)
- $\bar{m}_{k,t} = \frac{1}{N_{k,t}} \sum_{i:k_t^i = k} mom_{i,t}$ (average momentum in cluster k)
- $k_t^{(+)} = \arg\max_k \bar{m}_{k,t}$, $k_t^{(-)} = \arg\min_k \bar{m}_{k,t}$ (highest- and lowest-momentum clusters)
- $mom_{i,t}$: momentum of asset i
- $X_{i,t}$: momentum feature(s) for asset i; Z_t: momentum features for all other assets

Interpretation: $s_{i,t} = +1$ for assets in the highest-momentum cluster (long), -1 for assets in the lowest-momentum cluster (short), and 0 otherwise (no position).

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Motivating Gaussian Mixture Models for Momentum

Common application of Centroid Based, Soft Clustering

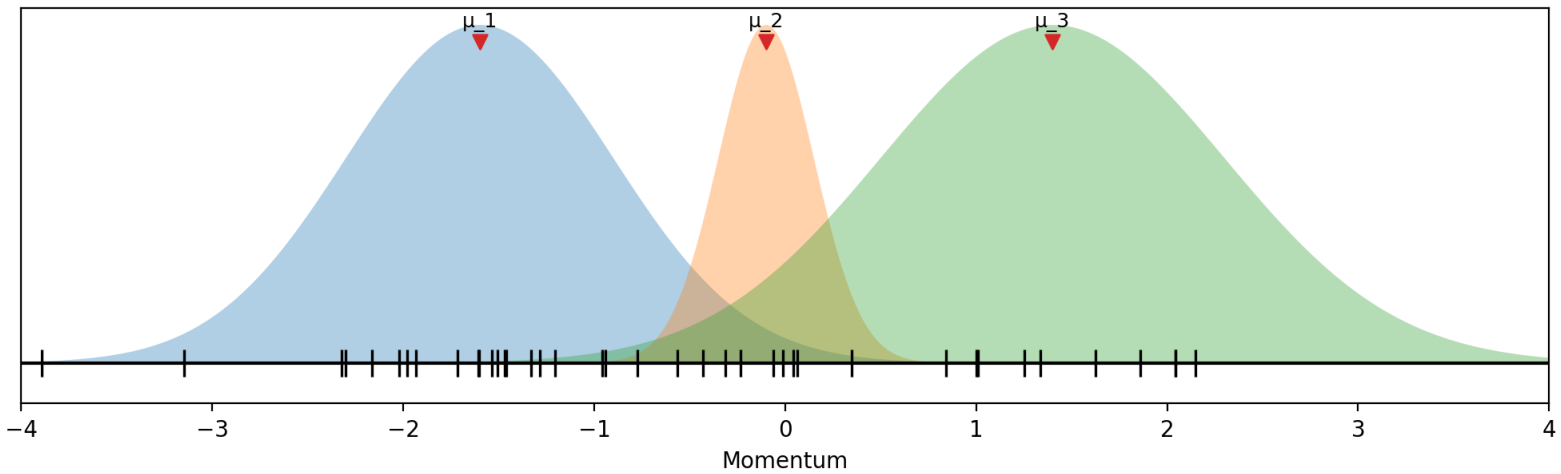
- Consider Momentum as a *continuous, noisy* signal
- There are no natural hard cutoffs between:
  - Losers
  - Neutral assets
  - Winners
- Assets near the boundary may plausibly belong to multiple regimes

GMM perspective:

- Observed momentum is generated by latent momentum regimes
- Each regime has its own mean and variance
- Each asset has a *probabilistic* cluster assignment

Motivating Gaussian Mixture Models (GMM)

Common application of Centroid Based, Soft Clustering



- Consider each bar on the horizontal axis to be an asset
- The asset near -4 is clearly in cluster 1
- The assets near 2 are clearly in cluster 3
- But the assets between -1 and 0.5 appear to have reasonable probabilities of being in multiple clusters

Steps for GMM Clustering

1. Initialize number of clusters K, cluster means $\bar{x}^k$, covariances Σ_x^k, and prior probabilities p^k that i is a member of cluster k
2. Update "degrees of belonging" (posterior probabilities)

$$\hat{p}(k^i = k \mid x_i) \propto \underbrace{\hat{p}^k}_{\text{prior}} \times \underbrace{N(x_i; \hat{\bar{x}}^k, \hat{\Sigma}^k)}_{\text{likelihood}}$$

3. For each cluster k update
   - Cluster probability: $\hat{p}^k = (N_k/N)$ with $N_k = \sum_{i=1}^N \hat{p}(k^i = k|x)$
   - Cluster mean: $\hat{\bar{x}}^k = (1/N_k) \sum_{i=1}^N \hat{p}(k^i = k|x) \times \mathbf{x}_i$
   - Cluster covar: $\hat{\Sigma}_x^k = (1/N_k) \sum_{i=1}^N \hat{p}(k^i = k|x) \times (\mathbf{x}_i - \hat{\bar{x}}^k)(\mathbf{x}_i - \hat{\bar{x}}^k)'$
   - Repeat until convergence

Output: predicted "degrees of belonging" $\hat{p}(k^i = k|x)$; i.e. probability that asset i is a member of each of the K clusters

GMM: Interpreting Posterior Probabilities

How does GMM determine an asset's degree of belonging?

- **Prior ($\hat{p}^k$):** How prevalent is cluster k before examining asset i?
- **Likelihood ($\mathcal{N}(x_i; \hat{\bar{x}}^k, \hat{\Sigma}^k)$):** How well does the asset's observed momentum fit the distribution of cluster k?
- **Posterior ($\hat{p}(k^i = k \mid x_i)$):** Probability of cluster membership after combining the prior and likelihood.

$$\hat{p}(k^i = k \mid x_i) = \frac{\hat{p}^k \, \mathcal{N}(x_i; \hat{\bar{x}}^k, \hat{\Sigma}^k)}{\sum_{j=1}^{K} \hat{p}^j \, \mathcal{N}(x_i; \hat{\bar{x}}^j, \hat{\Sigma}^j)}$$

Example: Consider an asset and two possible clusters.

	Prior	Likelihood	Product	Posterior
Cluster 1	0.80	0.10	0.08	0.40
Cluster 2	0.20	0.60	0.12	0.60

$$\hat{p}(k^i = 2 \mid x_i) = \frac{0.20(0.60)}{0.80(0.10) + 0.20(0.60)} = 0.60$$

Interpretation: Although cluster 2 is less prevalent, the asset's momentum is much more consistent with its distribution. GMM assigns a 60% probability of belonging to cluster 2.

Note: The likelihood is a probability density, not a probability. Posterior probabilities sum to 1 across clusters.

GMM Clustering Momentum

Q: How form a momentum signal via GMM Clustering?

1. Choose the number of clusters K via Elbow, Silhouette, etc.
2. Group assets by momentum via **GMM with a chosen threshold of cluster belonging** such that the cluster assignment for asset i at time t is the cluster for which the degree of belonging is the highest $k_t^i = k|max(\hat{p})$ to the cluster whose centroid is closest to its momentum value.

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$
$$p^{(+)} = \mathbf{1}\left[k_t^i = k_t^{(+)}\right], \qquad p^{(-)} = \mathbf{1}\left[k_t^i = k_t^{(-)}\right]$$

where

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Hierarchical Clustering

Hierarchical clustering groups data within a tree structure based on similarity

- Consider Momentum as a relative signal
- Assets may not be attractive or unattractive in isolation
- They are attractive relative to their peer group at a point in time

Hierarchical Clustering perspective:

- Arrange assets by similarity, not by arbitrary cutoff, as in K-Means
- Produce a ranking structure, not only clusters
- Choose the number of clusters *after* seeing the data ordering

Ways to create the hierarchy

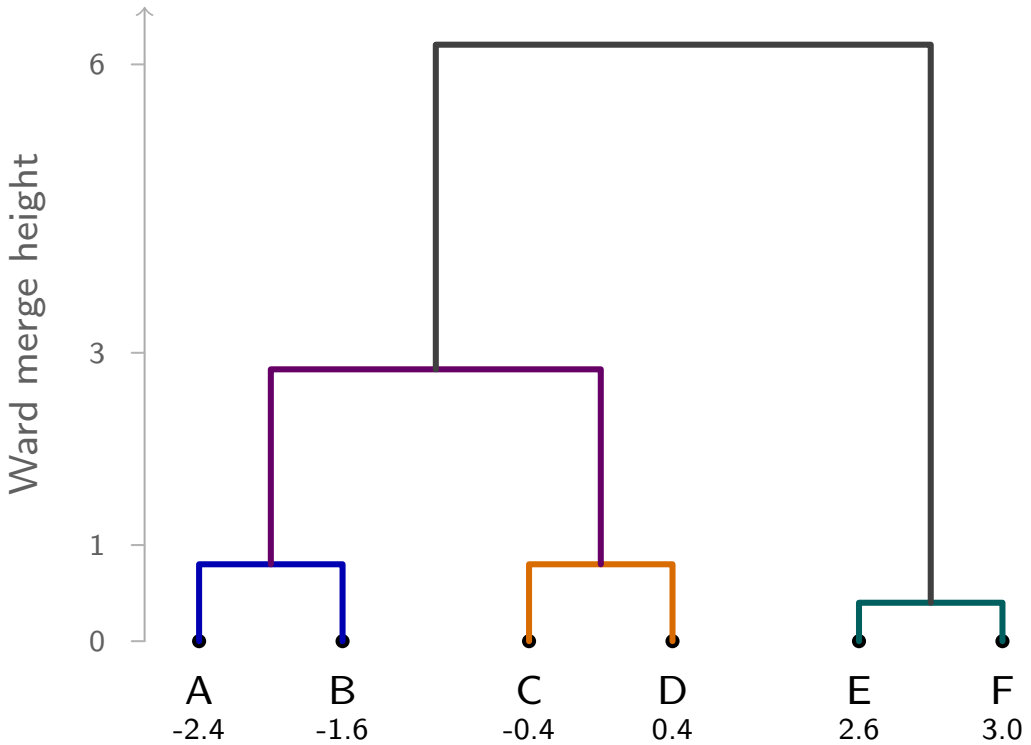
- Agglomerative: Start with each observation as a cluster and merge the closest pairs until a single cluster remains (typical)
- Divisive: Start with a single cluster and iteratively split into smaller clusters until each cluster contains only one observation

Steps for Agglomerative Clustering

1. Compute a pair-wise distance matrix of all observations e.g. $d(i,j) = (x_i - x_j)^2$
2. Initialize with one cluster per observation e.g. N clusters; one for each asset
3. Merge the two closest clusters based on either
   - Single linkage: distance = min pairwise distance between members
   - Complete linkage: distance = max pairwise distance between members
   - Average linkage: distance = average pairwise distance
   - Ward's method: merge clusters that minimize the increase in total within-cluster variance (This is typically a good default method because it tends to produce compact, relatively homogeneous clusters. Less appropriate if clusters have irregular shapes or very different sizes)
4. Update distance matrix to reflect the new cluster
5. Repeat until one cluster remains

Hierarchical Clustering: Building the Dendrogram

Six assets clustered by momentum using Ward's method



1. Start with six clusters, one per asset.
2. Merge E and F at height 0.40.
3. Merge A, B and C, D at height 0.80.
4. Merge AB with CD at height 2.83.
5. Finally, merge $ABCD$ with EF at height 6.21.

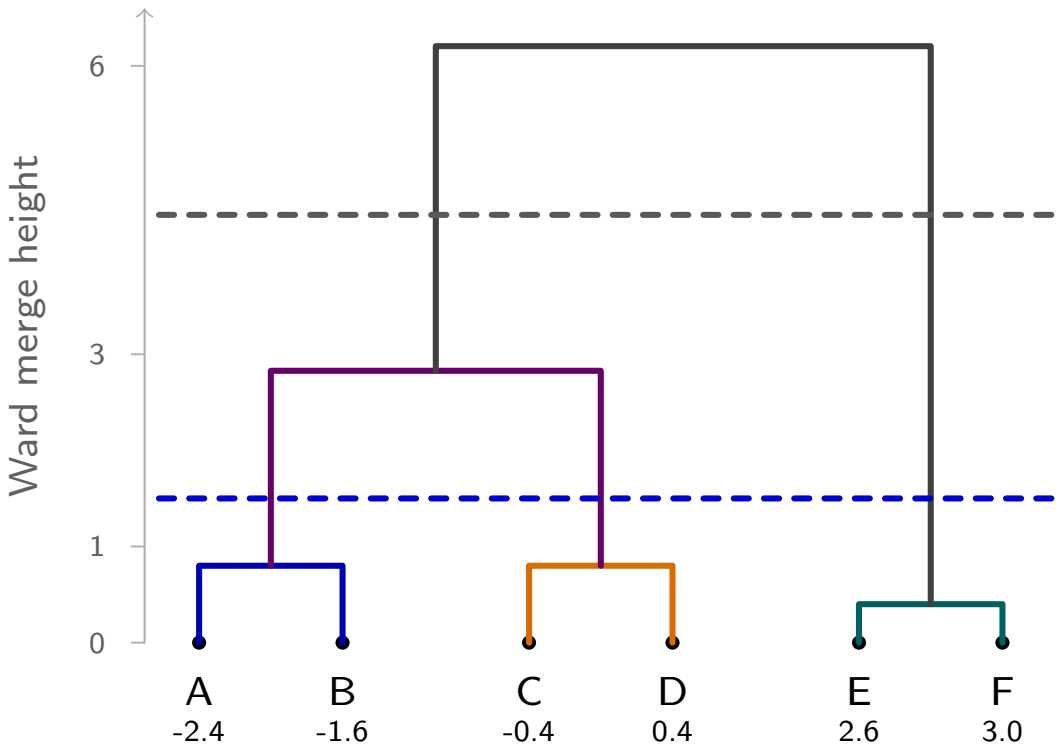
Why 2.83**?** $\bar{x}_{AB} = -2$, $\bar{x}_{CD} = 0$.

$$h = \sqrt{2\,\frac{2 \cdot 2}{2+2}\,(-2-0)^2} = \sqrt{8} \approx 2.83$$

Momentum values appear below the leaves; horizontal spacing is schematic. Ward height $h = \sqrt{2\,\Delta\mathrm{WCSS}}$.

Hierarchical Clustering: Interpreting the Dendrogram

Branches show merges; horizontal cuts determine the number of clusters



Q: Why does the *EF* branch extend so high?
A: *E* and *F* are close to one another, but their cluster stays separate until a much costlier merge.

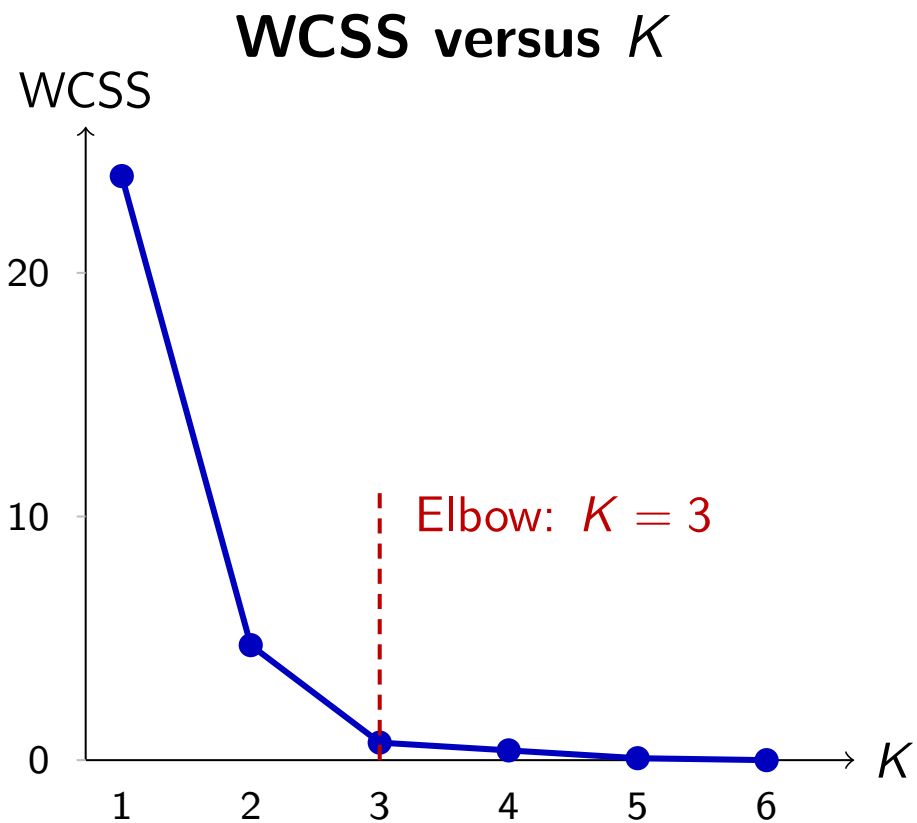
Q: How many clusters at the blue cut?
A: Three: *AB*, *CD*, and *EF*.

Q: How many at the gray cut?
A: Two: *ABCD* and *EF*.

Rule: A higher cut gives fewer, larger clusters; a lower cut gives more, smaller clusters.

Hierarchical Clustering: Choosing K

Cut height determines K; the trade-off is the same as in K-means



- **Dendrogram:** Look for large jumps in merge height; cut below a jump.
- **Elbow:** Cut the tree at each K, compute WCSS, and look for diminishing returns.
- **Silhouette:** Compare average silhouette scores across candidate values of K.

Here: $K = 3$ is plausible. Going from three clusters to two raises WCSS from 0.72 to 4.72.

Adapting Hierarchical Clustering for Multidimensional x

Suppose we are clustering on 12mth and 2mth returns such that $x_i = (12mth, 2mth)$

1. Compute a pair-wise distance matrix of all observations e.g., $d(i,j) = (1/2)[(12mth_i - 12mth_j)^2 + (2mth_i - 2mth_j)^2]$
2. Initialize with one cluster per observation e.g. N clusters; one for each asset
3. Merge the two closest clusters.
4. Update distance matrix to reflect the new cluster
5. Repeat until one cluster remains

Hierarchically Clustering Momentum

Q: How form a momentum signal via Hierarchical Clustering?

1. Choose the number of clusters K via Elbow, Silhouette, etc.
2. Assign each asset to a momentum cluster using hierarchical clustering with a chosen dissimilarity cutoff to the cluster whose centroid is closest to its momentum value.

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$
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The Network Perspective on Clustering

Graph-based clustering emphasizes relationships between observations

- Centroid-based methods group observations around representative points; GMM assumes underlying distributions.
- Alternatively, view each asset as a **node** in a network, with potential **edges** connecting pairs of assets.
- Pairwise similarity determines which assets are closely connected; groups may be linked to other groups by relatively weak connections.
- **MST perspective:** retain a sparse set of connections that links all assets with the smallest possible total distance. We can then identify clusters by removing weak links.

Measuring Similarity Between Assets

For return networks, transform correlations into distances

1. Compute historical returns and their pairwise correlations, ρ_{ij}.
2. Convert correlation into a distance:

$$d_{ij} = \sqrt{2(1 - \rho_{ij})} \in [0, 2].$$

ρ_{ij}	d_{ij}	Interpretation
1	0	Perfect positive co-movement
0.5	1	Moderate positive co-movement
0	$\sqrt{2} \approx 1.41$	No linear correlation
-1	2	Perfect negative co-movement

Interpretation: shorter distances indicate stronger positive return co-movement.

The transformation is a Euclidean distance between standardized return vectors when correlations use a common aligned return sample.

Constructing a Minimum Spanning Tree

Connect every asset while minimizing total edge distance

With N assets and pairwise distances d_{ij}, choose a set of edges E that minimizes

$$\min_{E} \sum_{(i,j)\in E} d_{ij} \qquad \text{s.t. the graph is connected and has no cycles.}$$

Prim's algorithm:

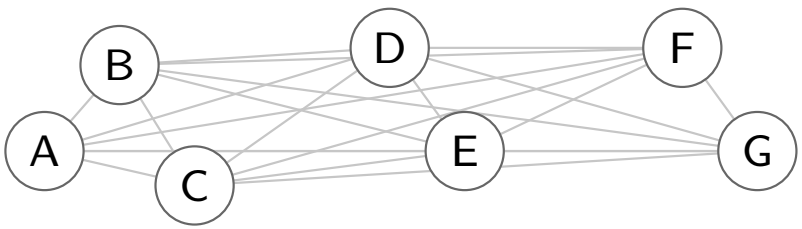
1. Start with any asset (one node in the tree).
2. Among all edges connecting a node *inside* the tree to a node *outside*, add the shortest one.
3. Repeat until all assets belong to the tree.

Output: one connected tree with exactly $N - 1$ edges. No cycles.

Visualizing a Minimum Spanning Tree

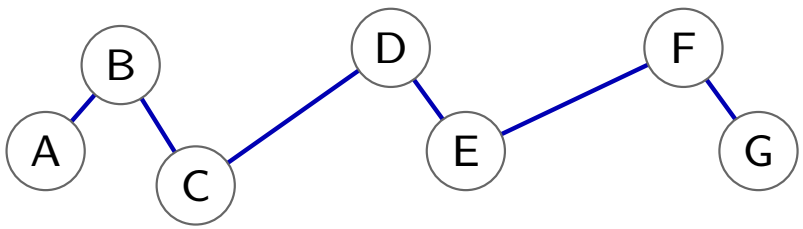
The MST retains only the connections needed to link all assets

All pairwise connections



Complete network: $\binom{7}{2} = 21$ possible edges

Minimum spanning tree



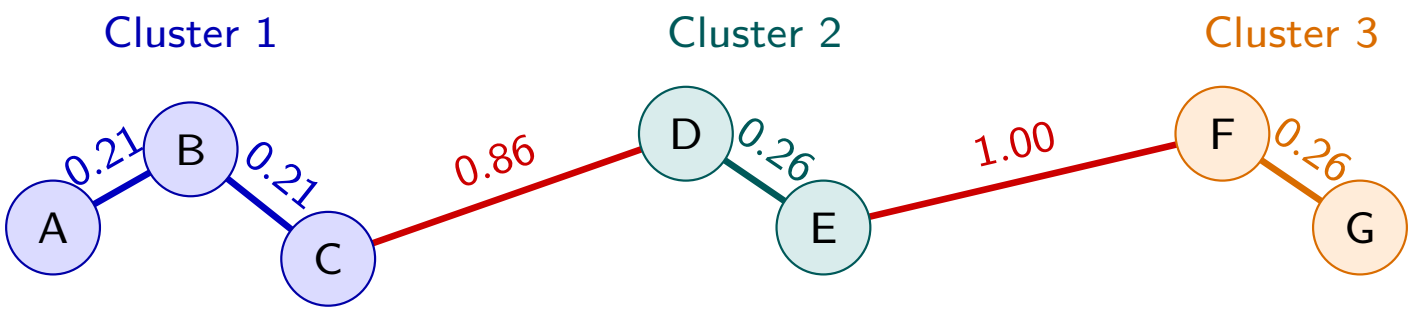
MST: $7 - 1 = 6$ edges, with minimum total distance

MST edge	*AB*	*BC*	*CD*	*DE*	*EF*	*FG*
Distance d_{ij}	0.21	0.21	0.86	0.26	1.00	0.26

Distances are illustrative and consistent (up to rounding) with the correlation-distance transformation. Node positions are schematic.

Clustering by Cutting the MST

Remove the $K-1$ largest edges to produce K connected groups



For $K = 3$, identify the two longest MST edges: EF (1.00) and CD (0.86). Cutting EF and CD gives a **forest** of three connected components: $\{A, B, C\}$, $\{D, E\}$, and $\{F, G\}$.

Choosing the Number of MST Clusters

Look for unusually weak links between otherwise close groups

Example: Sort the six MST edges from longest to shortest.

Rank	1	2	3	4	5	6
Edge	*EF*	*CD*	*DE*	*FG*	*AB*	*BC*
Distance	1.00	0.86	0.26	0.26	0.21	0.21

- The largest gap is between the second- and third-longest edges ($0.86 - 0.26 = 0.60$).
- This suggests removing the **two longest edges**, *EF* and *CD*, which are unusually weak connections.
- Removing two edges divides the MST into **three connected components**:

$$\{A, B, C\}, \qquad \{D, E\}, \qquad \{F, G\}.$$

- In general, removing $K - 1$ edges produces K clusters. Hence, $K = 3$.
- Validate candidate K using silhouette scores and cluster stability across time.

Caution: Edge gaps are heuristics, not a formal guarantee of the optimal K.

MST Clustering Momentum

Q: How form a momentum signal from correlation-based MST clusters?

1. Using a trailing return window, form the correlation-distance matrix and MST; cut its $K-1$ longest edges.
2. Compute each connected component's *average momentum*; go long the highest-momentum group and short the lowest.

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

$$p^{(+)}(X_{i,t}, Z_t) = \mathbf{1}[k_t^i = k_t^{(+)}], \qquad p^{(-)}(X_{i,t}, Z_t) = \mathbf{1}[k_t^i = k_t^{(-)}]$$

where

- k_t^i: index of the **connected component** containing asset i after cutting the MST (*not* a nearest-centroid label).
- $\overline{mom}_{k,t} = \dfrac{1}{N_{k,t}} \displaystyle\sum_{j:k_t^j=k} mom_{j,t}$: mean momentum in cluster k.
- $k_t^{(+)} = \arg\max_k \overline{mom}_{k,t}$, $k_t^{(-)} = \arg\min_k \overline{mom}_{k,t}$: highest- and lowest-momentum groups.
- $mom_{i,t}$: momentum of asset i, estimated using information available at time t.
- $X_{i,t}$: trailing return history of asset i; Z_t: trailing return histories of the other assets.

Interpretation: $s_{i,t} = +1$ for assets in the highest-average-momentum correlation cluster (long), -1 for assets in the lowest (short), and 0 otherwise.

Measuring Centrality in an MST

An additional use of the network: identifying prominent assets

Let $a_{ij} = 1$ if assets i, j share an MST edge (zero otherwise). Set $w_{ij} = a_{ij} \exp(-d_{ij})$, so shorter edges have greater similarity weight.

1. **Degree centrality:** number of direct connections.

$$\text{Degree}_i = \sum_j a_{ij}.$$

   Assets can have very different degrees even though the entire MST is connected.

2. **Weighted degree (strength):** number *and* similarity of direct connections.

$$\text{Strength}_i = \sum_j w_{ij}.$$

3. **Eigenvector centrality:** connections to assets that are themselves central.

$$Wc = \lambda_{\max} c, \qquad c_i = \frac{1}{\lambda_{\max}} \sum_j w_{ij} c_j, \qquad \sum_i c_i = 1.$$

A high score indicates a prominent network position; by itself it does not establish financial fragility or causal influence.